

KUALA LUMPUR INTL, Malaysia

WMO: 486500

Lat: 2.746N

Lon: 101.710E

Elev: 21

StdP: 101.07

Time Zone: 8.00 (E08)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 22.8	(c) 23.0	(d) 19.0	(e) 13.8	(f) 29.0	(g) 19.9	(h) 14.7	(i) 28.6	(j) 6.5	(k) 27.9	(l) 5.4	(m) 28.3	(n) 1.7	(o) 10	(p) 0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.3	34.2	25.7	33.8	25.8	33.1	26.0	28.0	31.7	27.6	31.3	27.2	31.0	3.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.8	30.9	26.2	21.7	30.0	26.1	21.6	29.9	89.8	31.5	87.7	31.8	86.3	31.4	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.6	(b) 5.0	(c) 4.4	(e) 21.8	(f) 35.5	(g) 0.4	(h) 1.0	(i) 21.5	(j) 36.3	(k) 21.3	(l) 36.9	(m) 21.0	(n) 37.5	(o) 20.7	(p) 38.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			20.5	29.6	0.6	1.3	20.1	30.6	19.7	31.3	19.4	32.0	18.9	33.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.9	28.5	28.7	28.7	28.7	28.5	28.0	28.0	27.9	28.0	27.7	27.7
	DBStd	1.06	1.12	1.01	1.14	0.99	0.91	1.07	0.98	0.93	1.00	0.93	0.82	1.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6636	556	518	578	561	578	554	557	557	538	558	531	550
	CDD18.3	3594	297	284	320	311	320	304	299	299	288	299	281	292
	CDH23.3	37811	2932	3035	3434	3394	3646	3482	3285	3233	3028	3013	2585	2743
	CDH26.7	14036	1038	1219	1377	1342	1441	1374	1202	1176	1093	1065	818	891
Wind	WSAvg	1.8	2.0	2.0	1.8	1.7	1.8	1.8	1.9	2.0	1.8	1.7	1.6	1.7
Precipitation	PrecAvg	2237	150	114	157	169	169	144	175	182	209	253	285	215
	PrecMax	2718	269	272	307	387	374	236	282	327	291	377	466	382
	PrecMin	1764	29	0	75	22	70	48	87	113	114	174	153	118
	PrecStd	262	71	81	64	95	79	48	49	50	49	53	93	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.9	35.2	34.5	34.2	34.2	33.2	33.3	33.8	33.7	33.1	33.2
		MCWB	25.4	24.4	24.4	26.5	26.7	25.8	25.6	25.5	25.3	26.0	26.5	25.9
	2%	DB	33.1	33.9	34.0	33.8	33.2	33.2	32.2	32.8	32.9	32.8	32.1	32.3
		MCWB	25.2	25.0	25.4	26.5	26.8	26.2	25.8	25.7	25.6	26.0	26.4	25.9
	5%	DB	32.2	33.1	33.1	32.9	32.8	32.6	32.0	32.0	32.1	32.0	31.4	31.8
		MCWB	25.3	25.3	25.9	26.6	26.8	26.3	25.8	25.9	25.9	26.1	26.5	25.9
	10%	DB	31.2	32.1	32.1	32.0	32.0	31.9	31.2	31.1	31.2	31.2	30.8	30.9
		MCWB	25.4	25.5	26.0	26.6	26.6	26.4	26.0	25.9	26.1	26.2	26.4	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	28.1	27.9	28.2	28.5	27.9	27.4	27.9	27.8	27.7	28.1	27.5
		MCDB	31.3	31.9	31.9	32.1	32.1	31.8	31.0	30.8	30.6	31.1	31.5	31.2
	2%	WB	26.7	27.1	27.2	27.7	27.7	27.4	27.0	27.1	27.0	27.0	27.2	26.7
		MCDB	30.6	31.2	31.3	31.6	31.5	31.3	30.6	30.6	30.6	30.8	30.7	30.4
	5%	WB	26.3	26.6	26.7	27.2	27.3	27.0	26.6	26.6	26.6	26.6	26.7	26.4
		MCDB	30.1	30.6	30.8	31.1	31.0	30.8	30.2	30.2	30.2	30.3	30.1	30.0
	10%	WB	25.9	26.1	26.4	26.8	26.9	26.6	26.3	26.2	26.2	26.3	26.4	26.1
		MCDB	29.6	30.2	30.4	30.5	30.6	30.3	29.8	29.8	29.8	29.9	29.6	29.5
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	8.0	7.6	7.3	7.0	6.9	7.0	7.2	7.3	7.1	7.0
		MCDBR	8.5	9.2	8.9	8.3	8.0	7.8	7.5	7.7	8.0	8.0	7.7	8.0
		MCWBR	3.4	3.8	3.5	3.4	3.2	3.2	3.2	3.2	3.3	3.2	3.3	3.4
	5% WB	MCDBR	7.8	8.5	8.0	7.7	7.4	7.1	6.9	7.1	7.2	7.4	7.2	7.3
		MCWBR	3.5	3.9	3.5	3.5	3.4	3.3	3.3	3.4	3.4	3.3	3.4	3.4
Clear-Sky Solar Irradiance	taub	0.472	0.515	0.523	0.507	0.487	0.522	0.509	0.537	0.550	0.575	0.481	0.457	
	taud	2.271	2.126	2.124	2.172	2.236	2.134	2.173	2.083	2.044	1.965	2.252	2.328	
	Ebn at Noon	856	826	816	812	807	766	783	778	782	766	841	864	
	Edh at Noon	140	164	165	153	139	152	147	165	176	190	141	130	
All-Sky Solar Radiation	RadAvg	4.65	5.37	5.39	5.25	5.00	4.84	4.77	4.87	4.78	4.75	4.55	4.30	
	RadStd	0.43	0.42	0.35	0.21	0.20	0.20	0.23	0.21	0.35	0.38	0.25	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.38	+0.21	N/A	+0.46	+0.41	+0.38	N/A	N/A	+137	+137

Nomenclature: See separate page